Version 1.75

Version 1.75 of the NO2 calibration takes a different approach to generating the finalized dataset involving stitching two different datasets together.

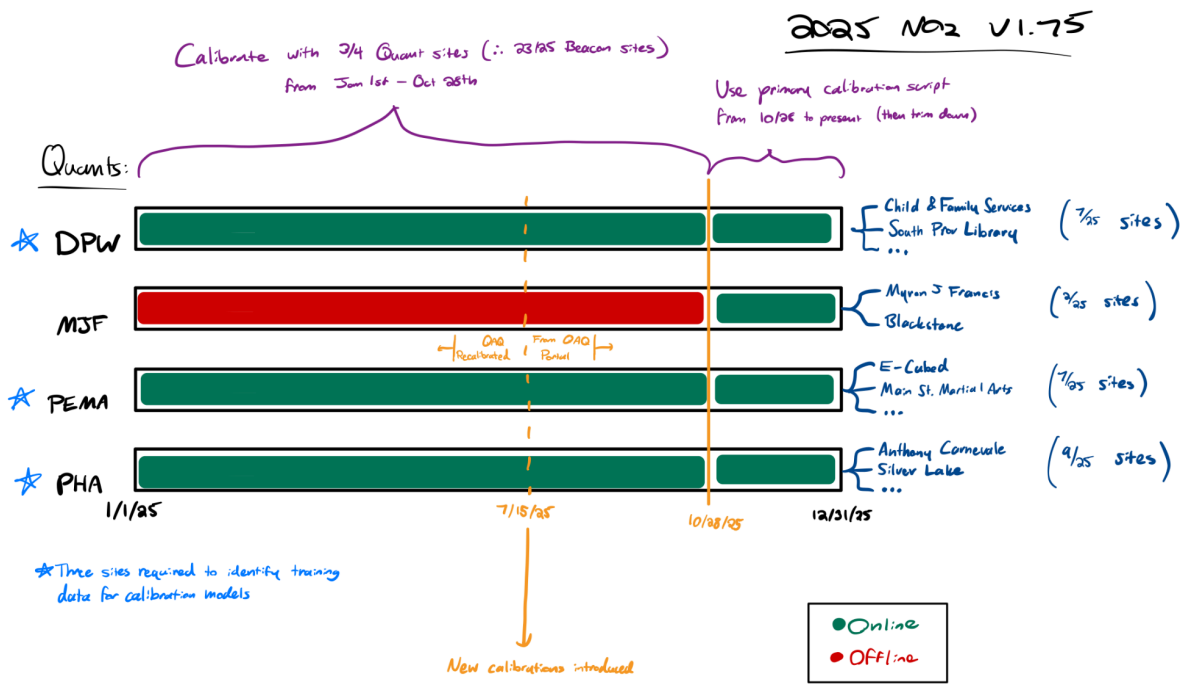
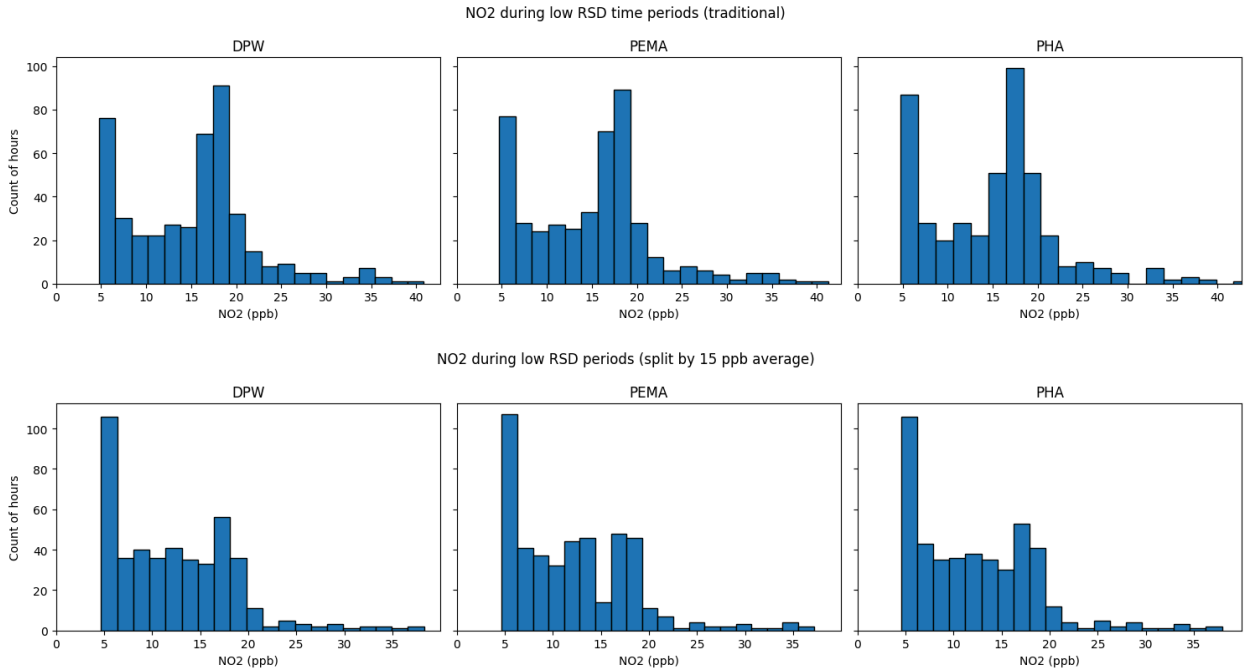


Figure that depicts the stitch process that is occurring during the 2025 NO2 v1.75. The major difference between this and version 1.5 is that two separate datasets are calibrated here.

Here, before 10/28/2025, 23 of 25 sites are calibrated separately from the rest of the network, simply using the three available Quants to identify low RSD time periods, then calibrating BEACO2Ns to those sites. This data is calibrated with a separate script, and the GitHub link is located here.

One process to note is that the "data split" that is used in the main NO2 calibration script for ~11/2025 and onward is not used here. The data split is used in the main v1 script to ensure that higher concentration values are represented in the training data while sacrificing some spatial homogeneity during low RSD time periods. This vastly increases collocated sensor performance, since the 5-15ppb range often doesn't end up being represented in the training data without the split. However, for the 2025 data, analysis showed that model performance decreased when including the data split, as it ended up over representing low concentration values.



Here, these graphs represent our training data values. The concentration distribution on the top (which doesn't use the data split) is better for training the model rather than the bottom distribution (which uses the data split) that overrepresents lower concentration values. Thus we do not "split" our training data like we do for 2026 onward.

Data from 10/28/25 onward is downloaded from the main version 1 script. This is then stitched in with data prior to 10/28/25 into one continuous dataset, and is trimmed down to only contain 2025 data. This generally produces more successful results based on collocated performance as compared to v1.5.

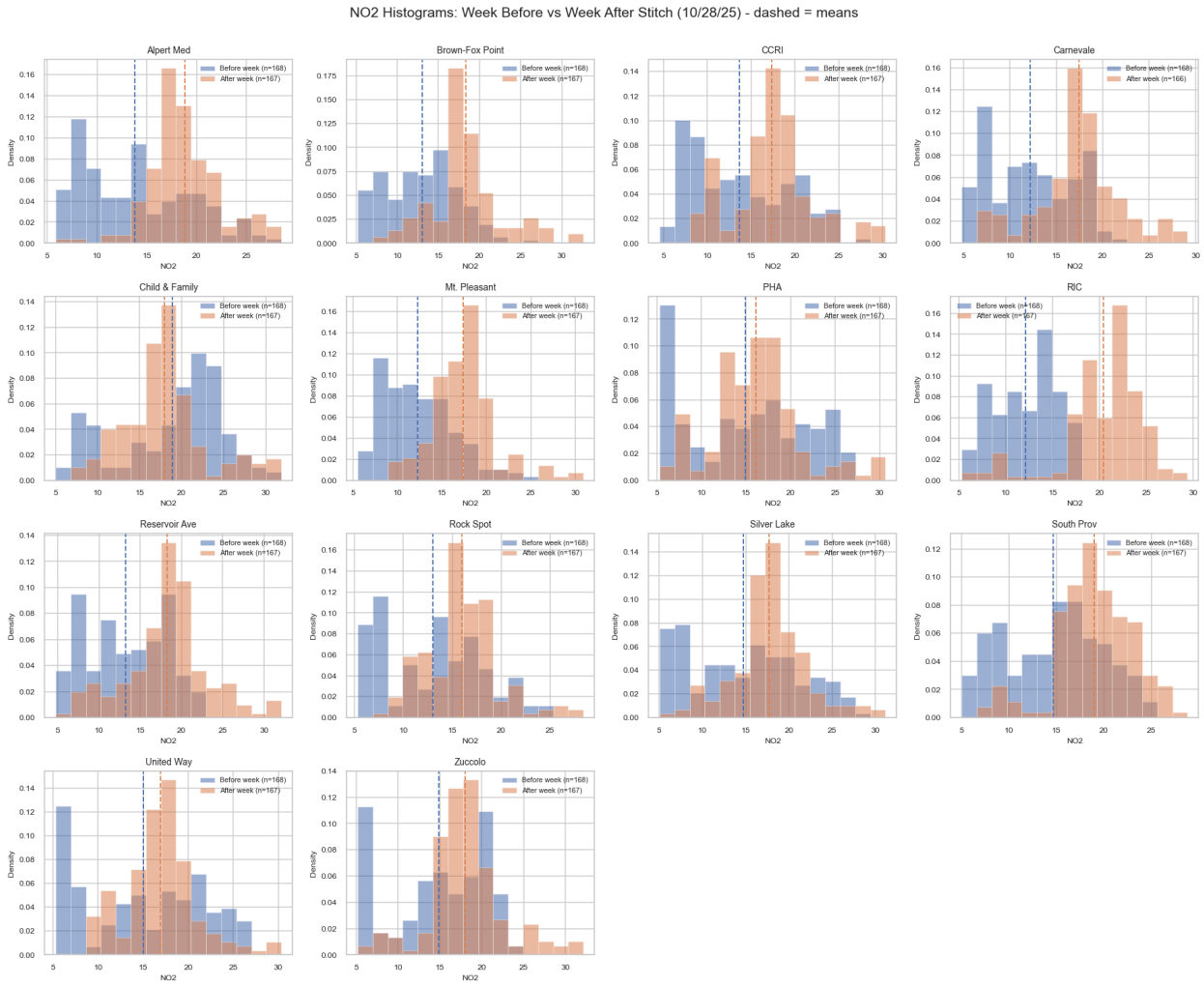
Site Name	Pre Stitch Test R^2	Pre Stitch Test RMSE	Post Stitch Test R^2	Post Stitch Test RMSE	Total R^2	Total RMSE
DPW	0.709	3.26	0.496	3.259	0.741	3.259
MJF	N/A	N/A	0.817	2.277	0.817	2.277
PEMA	0.826	2.705	0.374	4.374	0.757	3.178
PHA	0.783	3.11	0.875	1.979	0.825	2.922

Here "stitch" is the time stitch on 10/28/2025. The total R^2 and total RMSE represent the values for the entire 2025 time series that is available (i.e., MJF is still only late October-end of year).

The performance over the entire year is quite good at sites like MJF and PEMA, and is generally good at DPW and PEMA. I cannot account for the poor PEMA performance

post-temporal-stitch, but I have a feeling that seasonality plays a role, as temperatures were likely much lower and concentrations were higher.

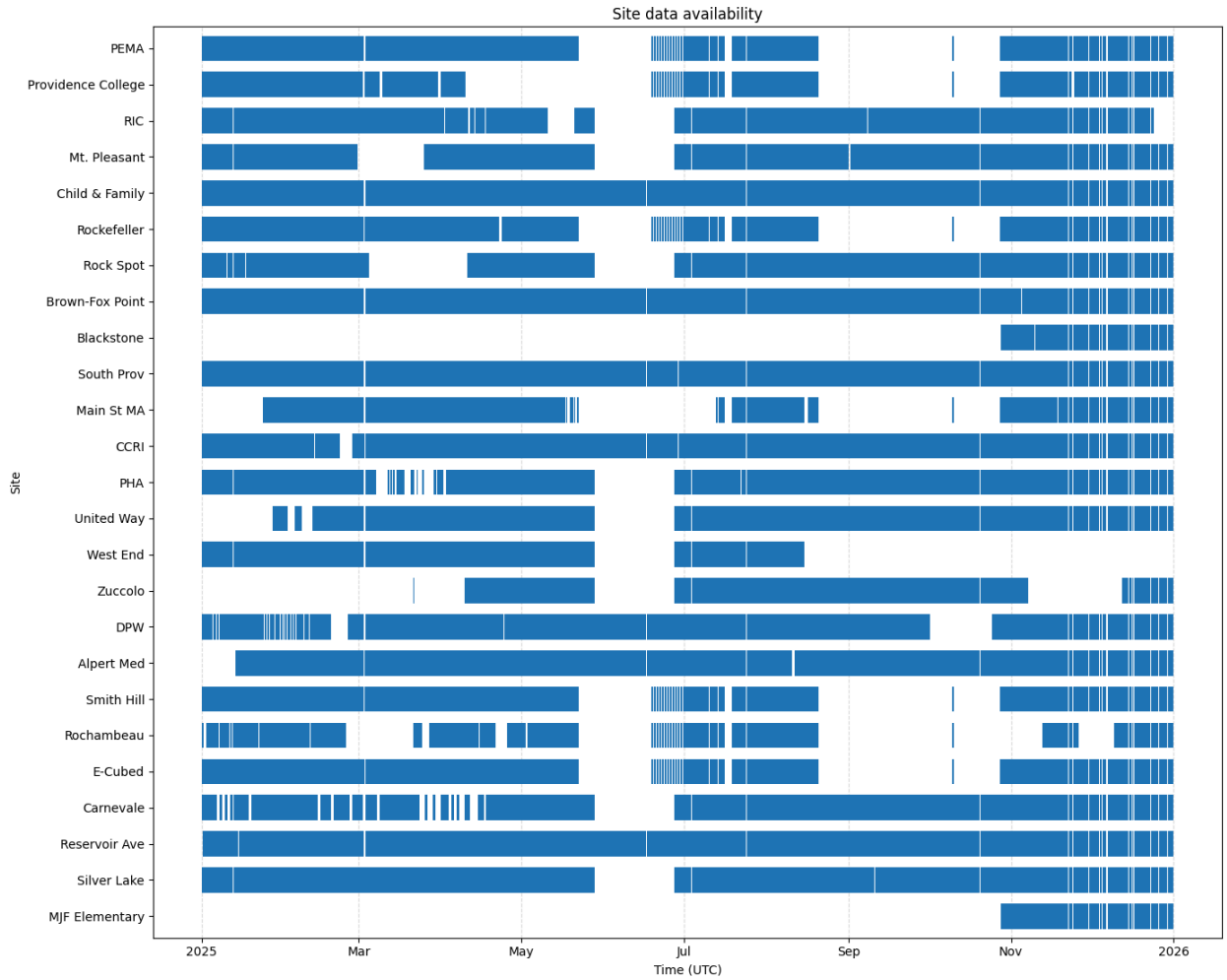
The only real potential issue with this data is the shift in concentration that occurs after the split. When looking at the week before and after the stitch, there seems to be a shift to a higher concentration regime:



All sites shift increase in median by ~4 ppb from the week before the stitch to the week after.

This shift is slight and could potentially be explained by temperature effects and seasonality. Warmer, lower concentration values are more prevalent in pre-stitch training data, which also may have caused the concentrations to be lower.

The collocated performance is high enough that it is worth taking this approach. Unless a new calibration method is produced for data after 10/28/25, this is the best dataset we have for now.



Data availability of v1.75 data by site.

Unfortunately, data availability by site is rather low. This is usually due to the reference Quants being offline. Some experimentation may occur where models are applied to BEACO2N data where no reference Quants are online to bring back some of this data availability.